

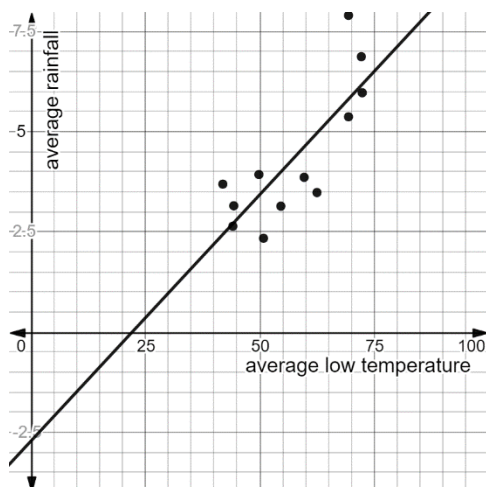
Algebra 1  
Unit 3  
Lesson 7 Homework

Name Answer key  
Date \_\_\_\_\_  
Period \_\_\_\_\_

A table shows the average low temperature and the average rainfall each month for a year in Jacksonville, Florida.

| Month                                  | Jan  | Feb  | Mar  | Apr  | May  | Jun  | July | Aug  | Sept | Oct  | Nov  | Dec  |
|----------------------------------------|------|------|------|------|------|------|------|------|------|------|------|------|
| <b>Average Low Temperature</b><br>(°F) | 41.9 | 44.3 | 49.8 | 54.6 | 62.5 | 69.4 | 72.4 | 72.2 | 69.4 | 59.7 | 50.8 | 44.1 |
| <b>Average Rainfall</b><br>(in)        | 3.69 | 3.15 | 3.93 | 3.14 | 3.48 | 5.37 | 5.97 | 6.87 | 7.9  | 3.86 | 2.34 | 2.64 |

The scatter plot shown represents the data from the table with a linear equation fit to model the equation.



This screenshot shows the slope,  $y$ -intercept, and correlation coefficient associated with the linear relationship modeled on the scatter plot.

$$y_1 \sim mx_1 + b$$

STATISTICS

$$r^2 = 0.6461$$

$$r = 0.8038$$

PARAMETERS

$$m = 0.122798 \quad b = -2.71046$$

RESIDUALS

$$e_1 \text{ plot}$$

- Write the equation for the line of best fit shown.

$$y = 0.122798x - 2.71046$$

2. Interpret the meaning of the slope in terms of the context.

As the average low temperature increases by  $1^\circ$ , the average rainfall increases by 0.122798 in.

3. Interpret the meaning of the  $y$ -intercept in terms of the context.

When the average low temperature is  $0^\circ\text{F}$ , the rainfall is  $-2.71046$  in. which is not possible.

4. Interpret the meaning of the correlation coefficient in terms of the context.

Strong, Positive, Linear

5. Use an inequality to express an appropriate domain in terms of the context.

Answers may vary.  
 $35 \leq x \leq 75$

6. Use an inequality to express an appropriate range in terms of the context.

Answers may vary.  
 $0 \leq y \leq 8$

7. Use the model to predict the average rainfall in a month with an average low temperature of  $74^\circ\text{F}$ .

$$y = 0.122798(74) - 2.71046$$
$$y \approx 6.4"$$

The average rainfall predicted would be 6.4in.

8. Use the model to predict the average rainfall in a month with an average low temperature of  $53^\circ\text{F}$ .

$$y = 0.122798(53) - 2.71046$$
$$y \approx 3.8\text{in.}$$

The predicted average rainfall would be 3.8 in.

9. Use the model to predict the average temperature in a month with an average rainfall of 2 inches.

$$2 = 0.122798x - 2.71046$$
$$x \approx 38.4^\circ$$

The predicted average temperature would be  $38.4^\circ$

10. Use the model to predict the average temperature in a month with an average rainfall of 6 inches.

$$6 = 0.122798x - 2.71046$$
$$x \approx 70.9^\circ$$

The predicted average temperature would be  $70.9^\circ$